

Overview of Tarot Consulting Chatbots Using Intelligence Artificial Intelligence

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Summary. In the context of Tarot chatbot systems using artificial intelligence developing rapidly since 2016, the lack of systematic research on technology, methods and evaluation standards has created a large gap in this field. This study conducts a comprehensive overview of Tarot AI consulting chatbot systems through analyzing 40 research documents from 2016 to 2025, including Vietnamese and English, using a systematic classification and comparison method according to five main groups including commercial products, open source code, datasets, technical approaches and evaluation methods. The analysis results show that three main technology trends include rule-based systems with the advantage of simplicity but lack of flexibility, traditional machine learning with better personalization capabilities but requiring large data, and large language models that dominate today with products such as Labyrinthos reaching more than one million downloads, Golden Thread with five hundred thousand users, and Sanctuary targeting the low-cost segment, mainly using GPT-4 and Claude Sonnet 3.5 with prompt engineering. Regarding evaluation methods, studies have proposed combining general indicators such as Coherence, Diversity, User Satisfaction with specific indicators such as Relevance to evaluate the relationship with the card, Hallucination Rate to check accuracy and Symbolic Depth to measure the depth of symbolic understanding. The study is structured into seven chapters: an introduction to the context and key terms, an overview of the open source and commercial product market, an analysis of the source code and technology framework, a review of existing Tarot data sets, a classification and comparison of research approaches, a synthesis of indicators and evaluation methods, and a conclusion on future development trends and research gaps that need to be filled.

Keywords: Tarot AI chatbot, large language model, prompt engineering, subjective AI assessment, spiritual consulting.

1 Introduction

Along with the strong development of major language models such as GPT-3, GPT-4 and Claude from 2019 onwards, the application of artificial intelligence in the field of spiritual consultation and personal reflection has begun to appear through automatic Tarot chatbot systems. Products such as TarotGPT, AI Tarot Reader or Discord Tarot Bot are widely deployed on many platforms, from web, mobile to messaging applications, to simulate

Traditional Tarot card spread process. However, as these systems are evolving, many technical and ethical challenges remain to be thoroughly addressed. Specifically, the illusion of large language models in generating interpretations, the lack of consistency across consultation sessions, and the psychological safety risks for users are urgent issues that need to be researched. This study conducts a comprehensive overview of Tarot consulting chatbot systems using artificial intelligence, from existing products, open source code, technical approaches, to datasets and evaluation methods proposed in previous studies. Thereby, the research provides a systematic view of the current state of technology, identifying key trends and gaps that need to be filled in the future. The research is presented through seven main sections. First is an introduction to the context, key terms and research structure. Next is an overview of Tarot AI products and applications currently on the market, including both commercial and open source products. The third part analyzes the source codes, frameworks and technologies used in different projects. The fourth section reviews existing Tarot data sets and data organization methods. The fifth part synthesizes and classifies research approaches from rule-based to LLM-based. The sixth part summarizes the indicators and evaluation methods proposed in the studies. Finally, there is a conclusion about development trends and future research directions.

Before going into detailed analysis, it is necessary to clarify some key terms and concepts used throughout this study to ensure readers have a unified understanding. Chatbots are computer programs designed to simulate a conversation with a user through a text or voice interface, using natural language processing and artificial intelligence techniques to understand user intent and provide appropriate responses (Vaswani et al., 2017). Tarot is a traditional symbolic system consisting of 78 cards divided into two main groups: 22 Major Arcana cards representing major life lessons and 56 Minor Arcana cards reflecting daily events, used for the purposes of reflection, self-awareness and advice. They should not be understood as absolute future prediction tools but as a means to support thinking and self-discovery (Song, 2019; Popat et al., 2019). Large language model or Large Language Model (LLM) is a type of artificial intelligence model trained on huge amounts of text data from billions to trillions of words, capable of understanding complex contexts and generating natural text like humans, with typical representatives such as OpenAI's GPT-4, Anthropic's Claude, and Google's Gemini (Sato, 2025; Elgammal et al., 2017). Prompt engineering is the technique of designing and optimizing input statements for LLM to achieve desired results, including many strategies such as role prompt-ing that requires the model to play the role of a specific character, few-shot learning that provides exemplars for the model to learn from, and chain-of-thought reasoning that guides the model to reason step by step (Ribeiro et al., 2016; Dixon, 2024). Illusion or hallucination in the context of AI is the phenomenon of a model generating false, inaccurate, or unrealistic information in training data, which is especially dangerous in consulting applications because it can give erroneous advice that affects the user's decisions (Sato, 2025; Bostrom and Yudkowsky, 2014). Regarding the approach, this study does not support considering Tarot as an absolute future prediction tool but rather sees it as a means to support reflection and self-development, so AI systems in this field need to be evaluated based on their ability to create conditions for users to self-discover and think.

reflection instead of prediction accuracy, a view supported by many modern studies on AI ethics (Floridi et al., 2018; Mittelstadt, 2019).

2 Product Market

Labyrinthos AI Tarot was founded in 2016, marking the pioneering start of the trend of applying artificial intelligence to the field of traditional Tarot card reading (Zhang, 2024). This application quickly attracted more than one million downloads, demonstrating the strong appeal of combining AI technology with classic Tarot art. Technically, Labyrinthos is built on the Software-as-a-Service platform with completely closed source code, protected as the company's intellectual property (Popat et al., 2019). The backend system is deployed on cloud services, ensuring security and stability during operation. The highlight of Labyrinthos lies in its comprehensive Tarot data set, built upon the traditional Rider-Waite-Smith deck but expanded with modern interpretations that reflect contemporary issues such as technology addiction, social media relationships, remote work, pandemic anxiety and the climate crisis (Treasure, 2025). The app is multilingual with translations for more than ten major languages to serve global users. In terms of features, Labyrinthos offers many spreads from basic to advanced such as Celtic Cross with ten positions, Three-Card Spread, Past-Present-Future spread, Horseshoe spread and Relationship spread (Dixon, 2024). The system integrates a large language model to create personalized interpretations based on the context of the user's question instead of extracting pre-existing text. Labyrinthos tracks interaction history to provide insight into recurring patterns, giving users insight into their personal journeys over time (Research Swinger Group, 2024). The business model combines basic free features and premium paid plans, free users have a limited number of views while premium plans allow unlimited access and exclusive features such as detailed historical storage, trend analysis and in-depth consulting. User requirements are minimal, requiring only a modern web browser with JavaScript enabled and a stable internet connection with a minimum bandwidth of one to two megabits per second (Ngo Ho, 2015).

Following the success of Labyrinthos, Golden Thread Tarot App emerged with over five hundred thousand active users, positioning itself as an attractive alternative for those looking for a Tarot experience that combines high aesthetics and advanced AI technology (Zhang, 2024). Golden Thread's main difference lies in its unique graphic design with a hand-drawn card deck illustration system in a modern art style, creating a unique visual experience compared to other applications that still use the traditional Rider-Waite-Smith image. Technically, Golden Thread adopts a three-tier architecture optimized for real production environments with a user interface layer built using React version eighteen with hooks to build dynamic interactive interfaces (Vaswani et al., 2017). The business logic layer combines a rule-based system written in Python with the FastAPI framework, which is responsible for analyzing user questions through natural language processing, randomizing cards according to the Fisher-Yates shuffle algorithm, determining card positions in spreads, and managing multi-turn conversation flows. The data layer uses MongoDB to store Tarot data, interaction history with long-term storage and

prompt templates for large language models (Ngo Ho, 2015). The outstanding strength is the integrated journaling feature that allows users to record thoughts and feelings after each reading, creating a personal spiritual diary to track their personal development journey. The app offers structured Tarot lessons through Labyrinthos Academy, an educational platform with basic to advanced courses for beginners (Song, 2019). Golden Thread integrates OpenAI's API with GPT-3.5-turbo or Claude 3 Sonnet to balance performance and cost (Elgammal et al., 2017). Regarding the pricing model, Golden Thread is positioned in a higher-end segment than Labyrinthos but is still competitive thanks to the added value of its exclusive artistic design and unique journaling feature (Chiea, 2024).

Sanctuary Tarot Bot appears as part of Sanctuary World, an integrated mental health care platform launched after Labyrinthos and Golden Thread, marking the trend of integrating Tarot AI into a large wellness ecosystem instead of a standalone application (Zhang, 2024). Sanctuary's uniqueness lies in its multi-service approach, with Tarot being just one of many mental health support tools including guided meditation, relaxation audio, psychological coaching and an online support community. In terms of technology, Sanctuary Tarot Bot is built on a chatbot platform that integrates with popular messaging platforms such as Facebook Messenger, Telegram and Discord through corresponding APIs, allowing users to access the Tarot card reading service directly from familiar chat applications without needing to install a separate app (Yamada-Rice & Dare, 2022). The system uses Python with the Flask framework for API endpoints and Discord.py for Discord bot integration, combined with MongoDB to store user data and interaction history. Sanctuary positions itself as the cheapest option among the top three apps, offering a freemium model with basic features completely free and a pre-mium package at a competitive price, suitable for young users with limited budgets (Dixon, 2024). A special feature is that Sanctuary integrates a community mechanism that allows users to share reading results, discuss interpretations and connect with people with similar interests in spirituality and personal development. Regarding AI integration, Sanctuary uses the free Google Gemini Pro API with rate limit for basic features, helping to save operating costs, while upgrading to more powerful models such as GPT-4 for premium users (Elgammal et al., 2017). This strategy allows Sanctuary to reach a wider public while maintaining service quality for the high-end customer segment (Fatfish Group Limited, 2024).

Tarot-AI is one of the first open source projects to appear on GitHub in 2020, marking the beginning of the movement to share knowledge about Tarot chatbots in the programmer community (Yamada-Rice & Dare, 2022). The project is released under the MIT License, allowing other developers to use it for free, modify it to their specific needs, and redeploy it in their own projects without paying copyright fees (Popat et al., 2019). Regarding architecture, Tarot-AI applies a basic three-layer architecture with the interface layer built using HTML5, CSS3 and vanilla JavaScript, the business logic layer using Python 3.8 with Flask 2.0 for lightweight API endpoints, and the data layer using SQLite for basic Tarot data storage (Ngo Ho, 2015). This project is basic, only providing a random card selection module with the Fisher-Yates shuffle algorithm and basic interpretation by taking text directly from the database without complicated AI generation. The source code capacity is about two to five megabytes including only the logical core, which is suitable

for beginners to learn how to build a Tarot chatbot. The installation method is detailed in the README.md file, requiring users to clone the repository from GitHub, set up a Python virtual environment, install dependent packages via pip install, and run the application locally to test on localhost port 5000 (Dixon, 2024). Although simple, the Tarot-AI project created the foundation for many more complex projects to come, making an important contribution to the dissemination of knowledge about Tarot AI chatbot development in the open source community.

TarotBot Discord appeared after Tarot-AI as an intermediate project, focusing on direct integration with the Discord platform through the Discord Bot API (Yamada-Rice & Dare, 2022). The project uses the Apache 2.0 License, which provides similar rights to the MIT License but has additional patent protection provisions, suitable for developers who want to protect their technical inventions (Popat et al., 2019). In terms of technology, TarotBot Discord is written in Python with the discord.py 2.0 library, a powerful async library for Discord bots with a cog system to organize code in a structured way. The business logic layer is upgraded compared to Tarot-AI basic, including conversation management through the state machine, supporting many popular card spreads such as Celtic Cross 10-card spread, Three-Card spread and Past-Present-Future spread (Dixon, 2024). The system uses PostgreSQL instead of SQLite to ensure better performance for production-grade applications when there are many concurrent users. A special feature is that TarotBot Discord integrates python-telegram-bot conversation handlers to manage multi-turn conversation flows, allowing the bot to understand the context of the question across multiple exchanges and provide coherent answers (Vaswani et al., 2017). Source code capacity increases to about ten to twenty megabytes due to additional dependencies and detailed documentation. The de-ploy method is clearly guided for cloud platforms like Heroku with limited free tier or Railway like modern alternative with generous free tier (Bostrom & Yudkowsky, 2014). This project is suitable for developers who want to create a Tarot chatbot that runs on their own Discord server.

Advanced Tarot Reader is the most advanced open source project, appearing in 2022 with full features of a commercial application (Dixon, 2024). This project is advanced, featuring context-based interpretation through prompt engineering with Large Language Models, user history tracking to provide insights into patterns in previous readings, a personalization engine that learns user preferences over time, and an analytics dashboard for developers to track usage metrics. In terms of technical architecture, Advanced Tarot Reader uses a modern technology stack with FastAPI 0.95 for high-performance API with automatic documentation, React 18 with hooks for the front-end to build its own web interface, and MongoDB for a flexible data layer (Ngo Ho, 2015). The most prominent feature is the integration with OpenAI API using GPT-3.5-turbo or GPT-4-turbo, allowing to create truly intelligent and personalized Tarot interpretations instead of just taking available text from the database (Elgammal et al., 2017). The project provides detailed prompt templates optimized for Tarot reading, including several-shot prompting examples, chain-of-thought reasoning, and role prompting for the LLM to act as a professional Tarot reader (Ribeiro et al., 2016). The project's JSON data set is very comprehensive with a schema that includes all fields such as card_id, name, type, suit, keywords_upright, keywords_reversed, meaning_upright, meaning_reversed and domain-specific interpretations for love, career, finance and spiritual, total file size from three to five.

megabytes (Song, 2019). The total source code capacity is up to twenty to fifty megabytes due to including front-end assets, dependencies and full documentation. Requirements from developers are quite high, needing intermediate-level programming knowledge in Python or JavaScript, understanding of RESTful API design principles, experience with Git version control system, familiarity with async programming concepts, understanding of natural language processing and knowledge of prompt engineering techniques (Ribeiro et al., 2016).

3 Source Code

Labyrinthos AI Tarot is a closed-source commercial product, the source code is not publicly available on any code sharing platforms such as GitHub, GitLab or SourceForge, and is protected as the company's intellectual property according to the Software-as-a-Service model (Popat et al., 2019). Users cannot download the source code to install the system themselves on a private server, but can only access it through the official web platform at labyrinthos.co or mobile applications on the App Store and Google Play Store. In terms of pricing, Labyrinthos adopts a freemium model with the free version offering basic features such as viewing cards a limited number of times per day, while the premium version requires payment of a monthly or annual subscription fee for unlimited access and advanced features (Zhang, 2024). Because there is no public source code, users absolutely cannot install this system on their own server or edit it according to personal needs. The tools needed from the end user side are simple, including only a modern web browser such as Chrome version 90 or higher, Firefox 88 or higher, Safari 14 or higher, or Edge 90 or higher with JavaScript enabled and cookies allowed to maintain session state, or an iOS/Android mobile device to install applications from the official app store (Ngo Ho, 2015). A stable internet connection with a minimum bandwidth of 1-2 Mbps is a necessary requirement for text-based interactions, as the system operates entirely in the cloud and requires continuous data transmission between the client and the server. Users do not need any programming tools or technical knowledge, just register an account via email or social network account and can start using it immediately through the intuitive graphical interface.

Golden Thread Tarot App is similar to Labyrinthos, also a commercial product with completely closed source code that is not publicly announced, developed and maintained by the Golden Thread company's own team (Zhang, 2024). There are no source code repositories related to this product on popular code sharing platforms, and the company applies strict protections to its intellectual property through terms of service that prohibit users from reverse engineering, data scraping or unauthorized access to the system. In terms of fee policy, Golden Thread is positioned in a slightly higher-end segment than Labyrinthos, offering a free trial version with limited features and a competitively priced premium package thanks to the added value of its exclusive artistic design and unique journaling feature that allows users to keep a spiritual diary (Chiea, 2024). There is no source code available for users to deploy themselves, all features are accessed through the official app distributed via the App Store for iOS and Google Play for Android, or via the official website.

official website. The "installation" process for the end user is simply downloading the application from the app store corresponding to the device's operating system, or accessing the website directly, without any technical configuration steps. Necessary tools include a smartphone or tablet running iOS 13 or higher or Android 8.0 or higher with approximately 100-200 MB of free storage space for the mobile application, or a computer with a web browser that supports modern HTML5 and CSS3 standards to access the web version (Vaswani et al., 2017). Users need to create an account using email or log in via Google, Facebook or Apple ID account, and can start using it immediately after completing registration without special technical knowledge.

Sanctuary Tarot Bot is part of the Sanctuary World platform, which also operates under a closed-source model without publicizing the source code, but has the special feature of being integrated into popular messaging platforms such as Facebook Messenger, Telegram and Discord instead of requiring a separate application installation (Zhang, 2024). The bot's source code is not posted on any public repository and is protected as the property of Sanctuary World. In terms of pricing model, Sanctuary positions itself as the cheapest option of the top three apps, offering a freemium model with completely free basic features including basic Tarot card reading and community participation, while the premium package has a significantly lower price than the competition, suitable for young users with limited budgets (Dixon, 2024). There is no public source code for users to host the bot on their own server, users can only access the bot through official channels provided by the company. The "installation" process in this case is even simpler than other applications, users just need to search for the name "Sanctuary Tarot Bot" on the Messenger, Telegram or Discord platform and start the conversation by sending the first message, the bot will automatically respond and instruct on how to use the features (Yamada-Rice & Dare, 2022). The only necessary tools are a pre-installed Messenger, Telegram or Discord application on your phone or computer, a valid user account on the respective platform, and an internet connection to exchange real-time messages with low latency. Users do not need to download or install anything additional, do not need to register a separate account for Sanctuary if they already have an account on the messaging platform, and can start reading Tarot cards immediately just by texting with the bot just like texting with a normal friend.

Different from the above commercial products, there currently exist a number of Tarot AI chatbot projects published open source on GitHub, GitLab and SourceForge, appearing mainly from 2020 onwards by individual programmers or small groups for the purpose of sharing knowledge and contributing to the community (Yamada-Rice & Dare, 2022). These open source projects are completely free, released under the MIT License or Apache 2.0 License allowing free use, copying, modification and distribution without paying copyright fees (Popat et al., 2019). Source code is available and public on repositories, users can download the entire source code to personal computers or private servers to research, edit and deploy according to needs. The process of installing these projects is much more complicated than using a commercial product, requiring users to perform many technical steps including: cloning the repository from GitHub using the git clone command, setting up a Python virtual environment using the command `python -m venv venv` or `conda create`, installing dependent packages via `pip install -r requirements.txt` for Python or `npm install` for Node.js, configuring environment variables in the `.env` file including the API

keys of OpenAI or Anthropic, initialize the database schema if needed, and run the application locally to test before deploying to the cloud (Ngo Ho, 2015). Necessary tools to install open source projects include: computer with Windows, macOS or Linux operating system, Git software to manage version control, Python version 3.8 or higher or Node.js version 16 or higher depending on the project language, text editor or IDE such as Visual Studio Code or PyCharm to edit code, terminal or command prompt to run installation commands, developer account on OpenAI or Anthropic to get API key if the project uses the this service, and basic programming knowledge of Python or JavaScript to be able to understand the code structure and make necessary adjustments (Ri-beiro et al., 2016). Although more complex, open source projects bring great benefits to the developer community who want to learn, research or build their own products based on available platforms without being bound by commercial terms.

4 Data

Tarot data plays a fundamental role in the development of Tarot consulting chatbot systems using artificial intelligence, including data about card meanings, spread structure, and various contextual interpretations (Dixon, 2024). The Rider-Waite-Smith deck, first published in 1909, is the most popular data source used in modern AI Tarot systems, with a standard structure of 78 cards divided into 22 Major Arcana cards representing major life lessons and 56 Minor Arcana cards reflecting daily events, divided into four suits of Wands, Cups, Swords and Pentacles corresponding to the elements Fire, Water, Air and Earth. (Popat et al., 2019). The basic data of this deck includes card names in English and other languages, digitized symbol images at high resolution from 300 to 600 DPI, main keywords usually from three to five words describing the spirit of the card, upright meaning when the card appears in its normal position and reverse meaning when the card is turned 180 degrees upside down, along with detailed interpretations according to specific fields such as sentiment, analysis of relationships, career. work-related, financial in terms of money management, and spirituality reflect the journey of inner development (Zhang, 2024). Labyrinthos AI Tarot uses an extensive data set based on Rider-Waite-Smith with modern interpretations rewritten to reflect contemporary issues of 21st century society such as technology and social media addiction, dating app relationships, remote work and the gig economy, pandemic anxiety and mental health, and the climate crisis and environmental consciousness, with data organized as a relational database with a card table containing underlying information. table, the meanings table stores contextual interpretations, the spreads table defines spread types, the positions table describes the meaning of each position, and the relationships table records the relationships between cards (Treasure, 2025; Song, 2019).

Golden Thread Tarot App develops its own data set with a flexible JSON structured classification system, including fields such as card_id, name, suit, element, keywords_upright, keywords_reversed, meaning_general from 200 to 300 words long, meaning_ing_love specializes in emotional questions, meaning_career focuses on career, meaning_finance analyzes financial aspects, meaning_spiritual refers to spiritual development, and visual_symbolism describes in detail the symbols in the card image (Song,

2019). The total JSON file size is about three to five megabytes for all 78 cards with full explanations, stored on MongoDB to support flexible queries and easily expand the schema by adding new fields (Elgammal et al., 2017). Sanctuary Tarot Bot uses a simpler data structure, focusing on core interpretations with a concise length optimized for the message format, limiting each interpretation to approximately 150 to 200 words divided into two to three short paragraphs, using simple, easy-to-understand language avoiding complex jargon (Yamada-Rice & Dare, 2022). Open source projects on GitHub provide many free Tarot data sets with different structures and qualities, in which the basic Tarot-AI project provides a simple CSV file with only three columns including card_name, meaning_upright, and meaning_reversed with a size of about 500KB, Ta-rotBot Discord uses PostgreSQL with a more complex schema including many tables standardized in third normal form, and Advanced Tarot Reader provides the most comprehensive JSON data set. with additional fields such as elemental_associations, astrolog-ical_correspondences, cultural_variations, historical_context and literary_references for a total capacity of up to 10MB (Dixon, 2024; Popat et al., 2019).

Table 1. Comparison of data structures of major Tarot AI systems

Characteristics	Labyri nthos	Golden Thread	Sanct uary	Open source code
Format	SQL	JSON + Mon- goDB	Simple JSON	CSV/JSON /SQLite
Number of leaves	78 full cards	78 full cards	78 full cards	22-78 leaves
Interpretation length	300- 500 words/ leaf	200-300 words/ leaf	150- 200 word s/leaf	100-300 words/leaf
Context Class:.....	5-7 fields	5 st an da rd ar ea s	2-3 fields	1-3 fields
Language support	10+ languag es	English mainly	Engli sh	English
Capacity	50- 100 MB	3-5 MB	1-2 MB	0.5-10 MB
Update	Quarte rly	Every year	Upon reque st	Uneven
Additional Metadata	Yes (abunda nt)	Yes (average)	No/lit tle	No/little

Regarding the method of collecting and building Tarot data, most modern AI systems rely on the manual process of digitizing traditional Tarot documents from printed books and reputable online sources, combined with inviting experienced Tarot experts to contribute to writing and quality checking the interpretations (Research Swinger

Group, 2024). Labyrinthos hires a team of 5 to 7 professional Tarot experts, each responsible for writing interpretations for a specific group of cards based on expertise, then all interpretations are compiled and go through at least three rounds of review by the editor-in-chief to ensure consistency, and finally beta tested with 100 to 200 users for 2 to 3 months to collect feedback before going into production (Zhang, 2024; Dixon, 2024). Golden Thread Tarot combines data from many classic sources such as "The Pictorial Key to the Tarot" by Arthur Edward Waite, "Seventy-Eight Degrees of Wisdom" by Rachel Pollack, then synthesizes to create a balanced interpretation between traditional wisdom and contemporary relevance standardized according to the JSON schema.

separately (Chiea, 2024; Song, 2019). Some open source projects initially experimented with web scraping to automatically collect data from public Tarot sites, but encountered many copyright issues, inconsistent quality, and required complex cleaning, so serious projects are now switching to manual curation methods that combine community contribution with a rigorous review process (Yamada-Rice & Dare, 2022; Popat et al., 2019).

The quality of Tarot data is evaluated through many criteria including accuracy compared to traditional sources, completeness of information covering all 78 cards with both upright and reversed meanings, consistency in terminology and style between cards, and extensibility to integrate new interpretations (Ribeiro et al., 2016). Commercial systems such as Labyrinthos and Golden Thread have rigorous quality control processes with multiple layers of testing including expert review, editorial review, technical review, user testing, and continuous monitoring after launch, using scoring systems to evaluate each interpretation according to dimensions such as accuracy, relevance, clarity, depth, empowerment and safety (Dixon, 2024; Research Swinger Group, 2024). A major challenge in standardizing Tarot data is the inherent polysemy of the symbols, where a card can be interpreted in many different ways depending on the school of Tarot, the context of the question, position in the spread, and relationship to other cards, raising the question of whether it is possible and advisable to standardize Tarot data too rigidly (Kim, 2021; Treasure, 2025).

A recent trend in Tarot data organization is to apply semantic knowledge graphs to represent complex relationships between cards, elements, planets, and other symbolic concepts (Kim, 2021). Advanced Tarot Reader tests using RDF (Resource Description Framework) to model Tarot knowledge in the form of triple subject-predicate-objects such as "The_Fool represents new_beginnings", "The_Tower opposes The_Emperor", forming a knowledge network that can be queried using SPARQL to answer complex questions (Song, 2019). Regarding training data for LLMs, most modern systems use prompt engineering with Tarot data embedded in the context window instead of fine-tuning due to flexibility and lower cost, although limited by context window sizes typically from 4K to 128K tokens (Vaswani et al., 2017; Sato, 2025). The Retrieval-Augmented Generation (RAG) method is emerging as an alternative solution, allowing to store Tarot knowledge in a vector database, retrieve top-k relevant pieces based on user query, and inject into LLM prompt, combining the advantages of both approaches in terms of scalability, updateability and efficiency (Lee et al., 2024; Ribeiro et al., 2016).

Table 2. Comparison of Tarot data integration methods with LLM

Method	Cost	Complexity	Update capabilities	Quality
Prompt Engineering	Low (\$0.03-0.06/1K tokens)	Low	Very easy (edit prompt)	Good for general cases
Fine tuning	Very high (\$500-5000/model)	High	Difficult (need re-train)	Best for specialized

RAG	Average (\$0.05- 0.10/query)	Average	Easy (more documents)	It's good to have high re-trieve accuracy
Hybrid (Prompt + RAG)	Average- Cao	High	Flexible	Best overall

Remaining challenges in Tarot data management for AI include copyright issues when digitizing content from published books, lack of standardization in data format and schema across different systems making interoperability difficult, need for continuous updates to reflect evolving interpretations as society changes, balancing traditional authenticity with modern relevance, and quality control at scale requiring sophisticated validation procedures (Dixon, 2024; Zhang, 2024). A number of community initiatives are attempting to build open standardized Tarot datasets such as the Tarot Data Commons initiative, Open Tarot Project, Linked Tarot Data, and Academic Tarot Corpus, but progress is slow due to limited resources, disagreements about scope, technical challenges and legal uncertainties (Treasure, 2025; Popat et al., 2019).

5 Research Approaches

In the field of Tarot consulting chatbots using artificial intelligence, research and application implementation from 2016 to present have applied three main technological approaches with different levels of complexity and effectiveness (Lee et al., 2019; Dixon, 2024). Analysis of research documents shows a clear shift from traditional rule-based methods to solutions using large language models, reflecting the trend of AI technology development in general and application to the field of spiritual consulting in particular (Zhang, 2024; Sato, 2025). Published studies on Tarot AI chatbots are classified into three main groups based on the core technology (Ribeiro et al., 2016; Song, 2019). The rule-based approach appeared as early as 2016, using pre-programmed rules to map directly from cards to interpretations available in the database (Yamada-Rice & Dare, 2022). The traditional machine learning approach, tested by several academic studies since 2018, applies classification algorithms to personalize user experiences (Lee et al., 2019; Song, 2019). The large language modeling approach is becoming mainstream in 2020, taking advantage of the natural text generation capabilities of models such as GPT-3, GPT-4, and Claude to create contextually appropriate Tarot interpretations (Zhang, 2024; Sato, 2025; Elgammal et al., 2017).

The rule-based approach was the earliest method used and is still popular in simple applications, especially basic open source projects on GitHub such as Tarot-AI (Yamada-Rice & Dare, 2022; Popat et al., 2019). In this approach, the system is designed to operate by mapping directly from the drawn card to the interpretive text available in the database through the use of fixed templates and decision trees to determine the appropriate answer (Dixon, 2024). Research by Popat et al. (2019) point out that the basic architecture of the rule-based system includes three main components: card database containing information about 78 Tarot cards, mapping rule engine

from card and position to interpretation template, and the text generation module combines the templates into a complete answer. The main advantages of this method highlighted by studies are simplicity in implementation that does not require in-depth knowledge of AI, low cost because there is no need for APIs of external AI services, the ability to completely control the answer content to ensure there is no false information, and high reliability because the output is always consistent with verified data (Dixon, 2024; Yamada-Rice & Dare, 2022). However, studies also point out significant limitations of rule-based systems including the lack of ability to personalize according to the context of a user's specific question, answers tend to be repetitive and unnatural due to the use of fixed templates, the inability to handle situations outside of pre-programmed scenarios, and limited scalability when wanting to add new features or improve the quality of interpretation (Ribeiro et al., 2016; Kim, 2021). Kim's (2021) study conducted a comparative evaluation showing that users rated rule-based systems as having a less natural and less satisfying experience than systems using AI text generation, with an average score of only 3.2 on a 7-point satisfaction scale. Lee et al. (2019) add that another problem with the rule-based approach is the rigidity of interpretation, the inability to create subtle variations in tone and style to suit different users, leading to a less personalized experience.

Traditional machine learning approaches have been tested by several academic studies to improve personalization compared to pure rule-based systems (Lee et al., 2019; Song, 2019). These methods often use algorithms such as Naive Bayes, Support Vector Machines, or Random Forest to classify user questions into categories and select appropriate interpretation templates, combined with collaborative filter-ing to learn user preferences from interaction history (Song, 2019). The study by Lee et al. (2019) proposed a hybrid architecture that combines a classification model to analyze user intent with a recommendation system to personalize interpretation, achieving 23% better personalization compared to a rule-based baseline in their experiments. Some more advanced systems are described by Song (2019) as applying sequence-to-sequence models or attention mechanisms to generate more personalized answers based on features extracted from the question and context. The advantages of this method shown by research include the ability to learn and improve from user data over time, better personalization than rule-based due to being able to learn personal preferences, lower operating costs compared to using commercial LLM APIs, and the ability to optimize for specific use cases (Elgammal et al., 2017; Lee et al., 2019). However, traditional ML systems still have many limitations that Ribeiro et al. (2016) analysis includes requiring large amounts of quality training data to achieve good performance, high complexity in developing and maintaining models, natural text generation is still limited compared to humans and LLMs, and there is a risk of overfitting or bias from training data if not carefully controlled. Research by Lustig and Rosner (2022) conducting user research with 50 participants shows that traditional ML systems achieve an average user satisfaction score of 4.1 on a scale of 7, higher than rule-based but still does not meet user expectations for naturalness and depth in interpretation.

The large language modeling approach is becoming a mainstream trend in modern Tarot AI applications, adopted by most successful commercial products.

public (Zhang, 2024; Dixon, 2024). These systems leverage the natural text generation capabilities of models such as OpenAI's GPT-3.5, GPT-4, Anthropic's Claude, and Google's Gemini to generate personalized and contextualized Tarot interpretations (Sato, 2025; Elgammal et al., 2017). According to the analysis of Lee et al. (2024), the LLM-based method has outstanding advantages including the ability to generate text naturally and coherently like humans, the ability to understand the complex and multi-dimensional context of the user question, no need to train the model from scratch but only need prompt engineering, the ability to handle diverse situations beyond pre-defined scenarios, and can be easily integrated via APIs of providers. However, this method also has its own challenges as warned by Bostrom and Yudkowsky (2014) such as high operating costs due to API call fees ranging from 0.03 to 0.06 USD per thousand tokens, risk of illusion when the model generates inaccurate information about card meanings, difficulty in fully controlling the content and tone of the answer, dependence on third-party services that may experience downtime or policy changes, and the need for mechanisms. multi-layered safety to avoid harmful content (Floridi et al., 2018; Mittelstadt, 2019).

Prompt engineering plays a key role in optimizing the performance of LLM-based systems, including many different strategies applied by research and applications (Ribeiro et al., 2016; Dixon, 2024). Role prompting is a technique that requires the LLM to act as a professional Tarot reader, helping to shape the tone and style of the answer, which Dixon (2024) has shown to significantly improve output quality compared to generic prompts. Few-shot prompts provide examples of good Tarot interpretations for LLMs to follow, helping to improve the quality and consistency of output according to research by Ribeiro et al. (2016). Chain-of-thought prompting guides LLM to make step-by-step inferences from card meaning to question context, described by Vaswani et al. (2017) pointed out to increase the logic and coherence of interpretation. Context injection puts complete information about the card drawn, position in the spread, and user question into the prompt, ensuring LLM has enough information to create an appropriate answer (Song, 2019). The Constraint specification clearly defines boundaries and limitations such as not making predictions that are too specific, not providing medical or legal advice, and always emphasizing that Tarot is a tool for reflection and not absolute prediction of the future, a requirement recommended by Floridi et al. (2018) emphasize responsible AI systems.

Sato's (2025) study on hallucination in LLMs shows that temperature control and top_p sampling have an important influence on the quality and stability of the Tarot AI system. Low temperatures in the range of 0.3 to 0.5 help increase consistency and reduce illusions but can make answers less creative and repeatable, suitable for cases where high precision is needed (Sato, 2025). The average temperature from 0.6 to 0.8 balances creativity and consistency, as reported by Lee et al. (2024) recommendations for most Tarot AI applications. High temperatures between 0.9 and 1.0 increase creativity and variety but have a higher risk of hallucinations and reduced coherence (Sato, 2025). Experimental studies show that the optimal configuration is usually temperature 0.7 combined with top_p 0.9, helping to achieve the best balance between factors (Lee et al., 2024; Dixon, 2024).

Regarding system architecture, modern Tarot AI applications often apply a three-layer architecture with a user interface layer, a business logic layer, and a data layer (Ngo Ho, 2015). The interface layer is built using modern web technologies such as React,

Vue.js or Angular for web apps, or React Native, Flutter for mobile apps, focuses on designing user-friendly and engaging experiences (Vaswani et al., 2017). The business logic layer often uses Python with FastAPI or Flask, or Node.js with Express, responsible for functions such as analyzing user questions through natural language processing, randomly selecting cards according to the Fisher-Yates shuffle algorithm, determining card positions in the spread pattern, creating appropriate prompts for LLM, calling the LLM provider's API, post-processing and filtering output, and managing session and conversation context (Dixon, 2024; Zhang, 2024). The data layer uses MongoDB for NoSQL flexibility, Post-greSQL for relational data, or Redis for caching and session management, storing Tarot data, user history, prompt templates and data analytics (Ngo Ho, 2015).

Safety and ethical mechanisms are integrated into the system through various layers of protection according to the framework proposed by Floridi et al. (2018) proposed. Input validation and filtering checks for and rejects inappropriate questions about serious medical, legal, self-harm, or harmful content (Bostrom & Yudkowsky, 2014). Prompt-level con-straints clearly define boundaries in the system prompt, require LLM to reject or redirect questions that cross boundaries, and always include an appropriate disclaimer (Mittelstadt, 2019). Output validation and post-processing use keyword filtering to detect harmful content, sentiment analysis to ensure appropriate tone, and fact-checking to reduce the illusion of card meaning (Chen et al., 2024). User education integrates disclaimers into all feedback, provides information about the limitations of AI, and encourages consultation with experts when needed (Wang et al., 2023). Monitoring and feedback collects user feedback on quality and safety, analyzes logs to detect patterns of problem cases, and continuously updates prompts and filters (Research Swinger Group, 2024).

Some pioneering studies have tested combined methods to take advantage of multiple approaches (Lustig & Rosner, 2022; Ribeiro et al., 2016). Retrieval-Augmented Generation combines rule-based retrieval with LLM generation, in which the system first retrieves relevant text from the Tarot database, then feeds it into LLM along with the question to create the final interpretation, helping to reduce illusions and improve accuracy (Ribeiro et al., 2016). Fine-tuning on domain data further trains the LLM base model on in-depth Tarot data, helping to improve the depth of symbol understanding and reduce inference costs, but requires large resources for data collection and training (Elgammal et al., 2017). Ensemble methods combine outputs from many different models or prompts, selecting or aggregating the best results, which increases reliability and reduces bias, but increases cost and latency (Goodfellow et al., 2014). Adaptive temperature scheduling adjusts the temperature based on the complexity of the question, using low temperatures for simple questions requiring high accuracy and higher temperatures for complex questions requiring creativity (Sato, 2025).

Regarding human-computer interaction, studies emphasize the importance of designing experiences that match Tarot users' expectations and needs (Lee et al., 2024; Yamada-Rice & Dare, 2022). Conversational design creates a natural conversation flow with the ability to follow-up questions, allowing users to clarify or deepen aspects, and maintain context across multiple exchanges. Personalization tracking and use of interaction history to recognize patterns in previous readings, adjust tone and style to suit user preferences, and provide insights into personal development trends over time (Treasure, 2025). Transparency and explainability explain how the system works

provide interpretation, show the user factors to consider such as cards, positions, context, and provide options to adjust parameters such as creativity level (Pasquale, 2015). Community features integrate the ability to share readings with the community, discuss and learn from others, and create connections between people with similar interests (Milan & Velden, 2023).

Recent studies also explore the integration of Tarot AI with other emerging technologies (Davies, 2024; Kim, 2021). Voice interface allows users to interact using voice instead of text, creating a more natural experience closer to a traditional Tarot reading, requiring integration of speech-to-text and text-to-speech technologies. Multi-modal input combines text, voice, and visual input such as photorealistic spreads, increasing the richness and flexibility of interactions (Duan et al., 2025). AR/VR applications create immersive experiences with virtual Tarot cards in 3D space, simulate the traditional reading space in a virtual environment, and combine visual storytelling with interpretation (Song, 2019). Cross-platform synchronization synchronizes data and history across multiple devices, allowing reading to start on mobile and continue on the web, and cloud data backup for accessibility from anywhere (Dixon, 2024).

The major challenge in this field was presented by Turing (1950) and Ribeiro et al. (2016) pointed out that it is a problem of evaluation and validation, due to the subjective nature of Tarot there is no ground truth for comparison. Studies recommending a variety of evaluation methods include comparative studies comparing AI with professional readers through blind tests, user studies collecting qualitative and quantitative feedback from actual users, technical metrics measuring coherence, relevance, diversity and hallucination rate, ethical audits checking compliance with AI ethics principles, and longitudinal studies tracking long-term impact on users. Kohavi et al. (2009) emphasize the importance of A/B testing and controlled experiments in continuously improving system quality, while Floridi et al. (2018) propose a comprehensive ethical framework for AI development in sensitive fields such as spiritual counseling.

6 Evaluation Indicators

Existing studies on Tarot AI chatbots show a trend of building multi-dimensional evaluation frameworks, combining quantitative and qualitative methods to measure system effectiveness in the context of this field being highly subjective and having no absolute “right answer” (Turing, 1950; Ribeiro et al., 2016). Literature review shows that evaluation frameworks are often formed based on five main groups of indicators including quality of generated content, user experience, safety and ethics, comparison with humans and technical indicators for practical implementation, including a combination of general indicators widely used in evaluating AI systems such as coherence, diversity, user satisfaction and service recommendation intention, along with specific indicators reflecting the characteristics of the problem Tarot AI such as the degree of connection between the interpretation and the card, the rate of misinterpretation, and the symbolic depth of the content (Floridi et al., 2018; Mittelstadt, 2019; Lee et al., 2024). A group of content quality indicators are used by studies to evaluate the level of logical consistency, contextual relevance, and the ability to generate diverse interpretations, thereby pointing out the core challenge in balancing

balance between consistency and creativity – a factor that is especially important for Tarot AI systems that rely heavily on symbolic interpretation and metaphorical language (Goodfellow et al., 2014; Ribeiro et al., 2016). In addition, the rate of erroneous interpretations is emphasized as an important specific indicator to detect cases where the system provides interpretations that are inconsistent with the card meaning or the context of the reading session, because these deviations can negatively affect the user experience and perception, especially when Tarot AI is used in the role of supporting personal reflection or consultation (Floridi et al., 2018; Sato, 2025). From a user perspective, comprehensive studies show that evaluating experiences does not stop at the level of immediate satisfaction but also extends to behavioral indicators such as repeat use and engagement, thereby reflecting the long-term perceived value and market acceptance of Tarot AI systems (Kohavi et al., 2009; Dixon, 2024). At the same time, safety and ethical issues are seen as integral components of current assessment frameworks, with a focus on the ability to control harmful content, maintain appropriate consultation boundaries, and avoid replacing important human decisions, indicating an increasing need for social responsibility when deploying AI in the field of spirituality and emotional support (Bostrom and Yudkowsky, 2014; Floridi et al., 2018). Comparative studies between AI Tarot chatbots and professional Tarot readers through blind tests were also compiled to clarify the degree of approach of AI compared to humans, in which the results show that AI achieves a significant level of similarity in some aspects such as personalization and usefulness, but there are still clear limitations in capturing the symbolic depth and subtle nuances of Tarot interpretation - experiential and cultural factors in which humans currently still dominate. (Ribeiro et al., 2016; Lee et al., 2024). Finally, technical indicators such as response time, operating cost, and system stability are included in the evaluation framework to ensure the feasibility of deploying Tarot AI in real environments, and demonstrate the trade-offs between response quality, processing speed, and cost that developers need to consider in accordance with specific application goals (Kohavi et al., 2009; Zhang, 2024). In general, current evaluation frameworks reflect the trend of an integrated approach, combining technical, experiential and ethical criteria, thereby providing a scientific basis for comprehensive evaluation of Tarot AI chatbot systems, while also pointing out research gaps related to measuring the symbolic depth and subjectivity of user experience in the context of generative AI.

7 Conclusion

This study conducted a comprehensive overview of Tarot consulting chatbot systems using artificial intelligence through systematic analysis of 40 research papers from 2016 to 2025, including academic studies and technical reports in Vietnamese and English. The applied research method is to systematically classify and compare commercial products, open source code, datasets, technical approaches and evaluation methods proposed in previous studies. The analysis results show three main technology trends in this field, including rule-based systems with the advantages of simplicity and low cost but lacking personalization capabilities.

and flexible, traditional machine learning with improved personalization but requires large amounts of training data and high complexity in development, and large language models are dominant today with successful products such as Labyrinthos reaching over a million downloads, Golden Thread with five hundred thousand users and Sanctuary targeting the low-cost segment, mainly using GPT-4 and Claude Sonnet with prompt engineering.

In terms of the product market and implementation technology, the analysis shows a clear stratification between different target groups with Labyrinthos positioned at the high end thanks to its comprehensive data set and modern interpretation that reflects contemporary issues such as technology addiction, social media relationships and pandemic anxiety, Golden Thread focusing on customers who appreciate aesthetics with unique graphic design and integrated journaling features, and Sanctuary targeting younger users with limited budgets. through a freemium model and integration with popular messaging platforms. The emergence of open source projects such as Tarot-AI, TarotBot Discord and Advanced Tarot Reader on GitHub since 2020 has made it easier for the developer community to access and research this technology, although the technical knowledge required is significantly higher than using commercial products. Analysis of the source code shows that commercial products adopt a closed-source model with the source code strictly protected as intellectual property, while open source projects released under the MIT License or Apache 2.0 License allow free use and modification, with most modern systems adopting a three-tier architecture using React or Vue.js for the frontend, Python with FastAPI or Flask for the backend, and MongoDB or Post-greSQL for the backend. database. database.

Evaluation of Tarot data sets and technical approaches shows that the Rid-er-Waite-Smith deck published in 1909 is still the most popular data source with a standard structure of 78 cards divided into 22 Major Arcana cards and 56 Minor Arcana cards, with different systems applying different organizational methods from Labyrinthos's relational database, Golden Thread's JSON with MongoDB, to optimization for Sanctuary's brief message. Recent trends in data organization are applying semantic knowledge structures and Retrieval-Augmented Generation methods that allow storing Tarot knowledge in vector databases. Regarding the technical approach, the analysis shows a technological shift from rule-based with a user satisfaction score of only 3.2 out of 7 points, to machine learning-ing-based reaching 4.1 out of 7, and now LLM-based is dominant with outstanding natural text generation capabilities. Prompt engineering plays a key role in optimizing the performance of LLM-based systems, including strategies such as role prompting, few-shot learning, chain-of-thought reasoning and con-straint specification, with the optimal configuration temperature 0.7 combined with top_p 0.9 to help balance creativity and consistency.

Analysis of evaluation methods and research contributions shows the trend of building a multi-dimensional evaluation framework that combines quantitative and qualitative methods, with the indicators used including a group of general indicators such as Coher-ence measuring coherence, Diversity evaluating diversity, User Satisfaction and Net Promoter Score measuring satisfaction and recommendation intention, combined with specific indicators such as Relevance evaluating the relationship between interpretation and cards, Hallucination Rate checking the ratio misinformation, and Symbolic Depth measures the depth of understanding of symbols. Comparative studies between AI and

Professional Tarot readers through blind tests show that AI is competitive in personalization but still significantly inferior in depth of symbolic understanding, reflecting the fact that current large language models cannot fully capture the deep layers of meaning that humans achieve through accumulated experience and intuition. The study contributes to the field by providing the first comprehensive overview of the Tarot AI chatbot ecosystem from commercial products to open source, systematically classifying three main technological approaches and analyzing the pros and cons of each, synthesizing prompt engineering techniques and optimal configurations for LLM-based systems, as well as analyzing ethical challenges and safety measures.

The study still has a number of limitations that need to be acknowledged and addressed in future studies, including that the scope of the review mainly focuses on systems based on the Rider-Waite-Smith deck and Western culture so generalizability to other Tarot systems or non-Western spiritual traditions requires further research, the rapidly evolving nature of AI technology especially large language models makes some performance findings potentially outdated, and most of the research The studies analyzed were technical reports or product descriptions lacking sufficient methodologically rigorous academic studies on long-term psychological effects. Future research directions include integrating Tarot AI with other spiritual consulting methods such as astrology, I Ching or numerology to create a more comprehensive consulting platform, developing support tools for professional Tarot readers rather than trying to completely replace them, long-term research on the psychological and behavioral effects of using Tarot AI chatbots, developing ethical principles and norms for the development and deployment of AI in the field of spiritual consulting, and Build a public standard data set for the research community. In summary, the field of Tarot consulting chatbots using artificial intelligence is in a period of rapid development with many opportunities and challenges, requiring a combination of technological innovation and ethical responsibility to ensure AI systems truly serve the best interests of users in their journeys of personal development and spiritual discovery.

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